Problem 1.1.1

PROMPT

Given the tensors

$$\boldsymbol{A} = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 0 & 1 & 3 \end{pmatrix} \text{ and } \boldsymbol{b} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

Solve:

SOLUTION

a) $\boldsymbol{A}_{ii}$

$$\boldsymbol{A}_{ii} \to \sum_{i}^{3} \boldsymbol{A}_{ii} = A_{11} + A_{22} + A_{33} = \text{trace}(A) = 8$$

b) $\boldsymbol{A}_{ij}\boldsymbol{b}_j$

$$c_i = A_{ij}b_j \to \sum_{j}^{3} A_{ij}b_j = A_{i1}b_1 + A_{i2}b_2 + A_{i3}b_3 =$$

$$= \begin{pmatrix} A_{11}b_1 + A_{12}b_2 + A_{13}b_3 \\ A_{21}b_1 + A_{22}b_2 + A_{23}b_3 \\ A_{31}b_1 + A_{32}b_2 + A_{33}b_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 6 \end{pmatrix}$$

c) $A_{ji}b_j$

$$c_i = A_{ji}b_j \to \sum_{j}^{3} A_{ji}b_j = A_{1i}b_1 + A_{2i}b_2 + A_{3i}b_3 =$$

$$= \begin{pmatrix} A_{11}b_1 + A_{21}b_2 + A_{31}b_3 \\ A_{12}b_1 + A_{22}b_2 + A_{32}b_3 \\ A_{13}b_1 + A_{23}b_2 + A_{33}b_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 6 \end{pmatrix}$$

Note: results for (b) and (c) are the same by coincidence

d) b_ib_j

$$A_{ij} = b_ib_j = \begin{pmatrix} b_1b_1 & b_1b_2 & b_1b_3 \\ b_2b_1 & b_2b_2 & b_2b_3 \\ b_3b_1 & b_3b_2 & b_3b_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & 4 \end{pmatrix}$$

e) $A_{ij}A_{ji}$

$$\alpha = \sum_{i}^{3}\sum_{j}^{3} A_{ij}A_{ji} =$$

$$
\begin{gathered}
A_{11}A_{11} + A_{12}A_{21} + A_{13}A_{31} + \\
A_{21}A_{12} + A_{22}A_{22} + A_{23}A_{32} + \\
A_{31}A_{13} + A_{32}A_{23} + A_{33}A_{33} \\
= 36
\end{gathered}
$$

Problem 1.1.2

PROMPT

Given the expressions in index notation, write the corresponding expressions in abstract tensor notation.

SOLUTION

a) $a_i b_i$

$$\alpha = a_i b_i = \sum_i^3 a_i b_i = \boldsymbol{a} \cdot \boldsymbol{b}$$

b) $a_i b_k c_k$

$$c_i = a_i \sum_k^3 b_k c_k \to \boldsymbol{c} = \boldsymbol{a} \cdot (\boldsymbol{b} \cdot \boldsymbol{c})$$

c) $A_{ik} B_{kj}$

$$C_{ij} = A_{ik} B_{kj} =$$

$$\sum_k^3 A_{ik} B_{kj} \to$$

$$\begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} + A_{13}B_{31} & A_{11}B_{12} + A_{12}B_{22} + A_{13}B_{32} & A_{11}B_{13} + A_{12}B_{23} + A_{13}B_{33} \\ A_{21}B_{11} + A_{22}B_{21} + A_{23}B_{31} & A_{21}B_{12} + A_{22}B_{22} + A_{23}B_{32} & A_{21}B_{13} + A_{22}B_{23} + A_{23}B_{33} \\ A_{31}B_{11} + A_{32}B_{21} + A_{33}B_{31} & A_{31}B_{12} + A_{32}B_{22} + A_{33}B_{32} & A_{31}B_{13} + A_{32}B_{23} + A_{33}B_{33} \end{pmatrix}$$

$$= \boldsymbol{A} \cdot \boldsymbol{B}$$

d) $A_{ki} B_{kl} A_{lj}$

Assuming the answer as D_{ij}, since only i and j are non repeating, assume

$$C_{il} = A_{ki} B_{kl} \to \boldsymbol{C} = \boldsymbol{A}^T \cdot \boldsymbol{B}$$

Then

$$D_{ij} = C_{il} A_{lj} \to \boldsymbol{D} = \boldsymbol{C} \cdot \boldsymbol{A}$$

Finally

$$\boldsymbol{D} = \boldsymbol{A}^T \cdot \boldsymbol{B} \cdot \boldsymbol{A}$$

Another approach is:

Assume

$$E_{kj} = B_{kl}A_{lj} \rightarrow \boldsymbol{E} = \boldsymbol{B} \cdot \boldsymbol{A}$$

Then

$$D_{ij} = A_{ki}E_{kj} \rightarrow \boldsymbol{D} = \boldsymbol{A}^T \cdot \boldsymbol{E}$$

Which, of course, results in the same value!

$$\boldsymbol{D} = \boldsymbol{A}^T \cdot \boldsymbol{B} \cdot \boldsymbol{A}$$

e) $a_i A_{jk} b_k$

Let the answer be a matrix D_{ij}.

Let $c_j = A_{jk}b_k$

$$c_j = A_{j1}b_1 + A_{j2}b_2 + A_{j3}b_3 \rightarrow \boldsymbol{c} = \boldsymbol{A} \cdot \boldsymbol{b}$$

then,

$$D_{ij} = a_i c_j \rightarrow \begin{pmatrix} a_1c_1 & a_1c_2 & a_1c_3 \\ a_2c_1 & a_2c_2 & a_2c_3 \\ a_3c_1 & a_3c_2 & a_3c_3 \end{pmatrix} = \boldsymbol{a} \cdot \boldsymbol{c}^T$$

Thus

$$\boldsymbol{D} = \boldsymbol{a} \cdot (\boldsymbol{A} \cdot \boldsymbol{b})^T$$

f) $a_i A_{ik} b_k$

From the previous question, we know that

$$c_i = A_{ik}b_k \rightarrow \boldsymbol{c} = \boldsymbol{A} \cdot \boldsymbol{b}$$

And, simply

$$\alpha = a_i c_i = \boldsymbol{a} \cdot \boldsymbol{c}$$

The answer is

$$\alpha = \boldsymbol{a} \cdot \boldsymbol{A} \cdot \boldsymbol{b}$$

Problem 1.2.1

PROMPT

Show that $\delta_{ij}\delta_{ji} = 3$.

SOLUTION

$$\delta_{ij}\delta_{ji} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^T = 1 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 = 3$$

Problem 1.2.2

PROMPT

Find the expression in index notation for

$$\frac{\partial \boldsymbol{v}}{\partial \boldsymbol{v}}$$

where $\boldsymbol{v}$ is a vector.

SOLUTION

$$\boldsymbol{A} = \frac{\partial \boldsymbol{v}}{\partial \boldsymbol{v}} = \begin{pmatrix} \frac{\partial v_1}{\partial v_1} & \frac{\partial v_1}{\partial v_2} & \frac{\partial v_1}{\partial v_3} \\ \frac{\partial v_2}{\partial v_1} & \frac{\partial v_2}{\partial v_2} & \frac{\partial v_2}{\partial v_3} \\ \frac{\partial v_3}{\partial v_1} & \frac{\partial v_3}{\partial v_2} & \frac{\partial v_3}{\partial v_3} \end{pmatrix} \to A_{ij} = \frac{\partial v_i}{\partial v_j}$$

It is known that

$$\frac{\partial v_l}{\partial v_m} = \begin{cases} 1 \text{ if } l = m \\ 0 \text{ else} \end{cases}$$

Then it is safe to assume that

$$\frac{\partial v_i}{\partial v_j} = \delta_{ij}$$

Problem 1.2.3

PROMPT

With the aid of the Kronecker delta, find the expression in index notation for the trace of $\boldsymbol{a}\boldsymbol{b}^T$ where $\boldsymbol{a}$ and $\boldsymbol{b}$ are vectors.

SOLUTION

$$\alpha = \text{trace}(\boldsymbol{a}\boldsymbol{b}^T) = a_i b_j \delta_{ij}$$

With the contraction property

$$b_j \delta_{ij} = b_i$$

$$\alpha = a_i \cdot b_i = \boldsymbol{a} \cdot \boldsymbol{b}$$

Problem 1.2.4

PROMPT

Simplify the expression

$$\frac{\partial^2}{\partial x_i \partial x_j}(x_i x_j)$$

SOLUTION

$$\frac{\partial^2}{\partial x_i \partial x_j}(x_i x_j) = \frac{\partial}{\partial x_i}\left(\frac{\partial x_i}{\partial x_j} x_j + \frac{\partial x_j}{\partial x_j} x_i\right) =$$

$$= \frac{\partial}{\partial x_i}(0 + \delta_{jj} x_i) = 3\frac{\partial}{\partial x_i}(x_i) = 3\delta_{ii} = 9$$

Problem 1.3.1

PROMPT

Show that

$$\epsilon_{ijk} a_i a_j = 0$$

SOLUTION

By brute force...

Let $A_{ij} = a_i a_j$

$$\left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}\right) \cdot \begin{pmatrix} a_1 a_1 & a_1 a_2 & a_1 a_3 \\ a_2 a_1 & a_2 a_2 & a_2 a_3 \\ a_3 a_1 & a_3 a_2 & a_3 a_3 \end{pmatrix} = \begin{pmatrix} +a_2 a_3 - a_3 a_2 \\ -a_1 a_3 + a_3 a_1 \\ +a_1 a_2 - a_2 a_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

By triviality

Multiplying both sides by the cartesian basis vector:

$$\epsilon_{ijk} a_i a_j \cdot \hat{\boldsymbol{e}}_i = 0 \cdot \hat{\boldsymbol{e}}_i = 0$$

The expression above is the cross product of the vector $\boldsymbol{a}$.

$$\epsilon_{ijk} a_i a_j \cdot \hat{\boldsymbol{e}}_i = \boldsymbol{a} \times \boldsymbol{a}$$

By definition, the cross product of a vector by itself is 0, since they are colinear.

Problem 1.3.2

PROMPT

Prove de identity

$$\epsilon_{ijk}\epsilon_{mnk} = \delta_{im}\delta_{jn} - \delta_{in}\delta_{jm}$$

SOLUTION

$$\epsilon_{ijk}\epsilon_{mnk}$$

The product of two Levi-Civita tensors can be written as the determinant of a 3×3 matrix of Kronecker deltas:

$$\epsilon_{ijk}\epsilon_{mnp} = \det \begin{pmatrix} \delta_{im} & \delta_{in} & \delta_{ip} \\ \delta_{jm} & \delta_{jn} & \delta_{jp} \\ \delta_{km} & \delta_{kn} & \delta_{kp} \end{pmatrix}$$

We aim to solve the specific case where $p = k$:

$$\epsilon_{ijk}\epsilon_{mnk} = \det \begin{pmatrix} \delta_{im} & \delta_{in} & \delta_{ik} \\ \delta_{jm} & \delta_{jn} & \delta_{jk} \\ \delta_{km} & \delta_{kn} & \delta_{kk} \end{pmatrix}$$

Expanding the determinant along the bottom row, and knowing that $\delta_{kk} = 3$ and $\delta_{ik}\delta_{km} = \delta_{im}$, the matrix simplifies directly to the identity:

$$\begin{aligned} \epsilon_{ijk}\epsilon_{mnk} &= \det \begin{pmatrix} \delta_{im} & \delta_{in} \\ \delta_{jm} & \delta_{jn} \end{pmatrix} \\ &= \delta_{im}\delta_{jn} - \delta_{in}\delta_{jm} \end{aligned}$$

Problem 1.3.3

PROMPT

Using the permutation operator, prove the vector triple product identity

$$\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c}) = \boldsymbol{b}(\boldsymbol{a} \cdot \boldsymbol{c}) - \boldsymbol{c}(\boldsymbol{a} \cdot \boldsymbol{b})$$

SOLUTION

Let $\boldsymbol{r}$ be the result vector.

$$\boldsymbol{r} = \boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c}) \to r_i = \epsilon_{ijk} a_j d_k$$

For $d_k = \epsilon_{kmn} b_m c_n$

Therefore

$$r_i = \epsilon_{ijk} a_j \epsilon_{kmn} b_m c_n$$

From the property obtained from the last problem, and the fact that $\epsilon_{kmn} = \epsilon_{mnk}$,we know that:

$$\epsilon_{ijk}\epsilon_{mnk} = \delta_{im}\delta_{jn} - \delta_{in}\delta_{jm}$$

$$\begin{aligned} r_i &= (\delta_{im}\delta_{jn} - \delta_{in}\delta_{jm}) a_j b_m c_n \\ &= a_j b_m c_n \delta_{im}\delta_{jn} - a_j b_m c_n \delta_{in}\delta_{jm} \\ &= b_i (a_j c_j) - c_i (a_j b_j) \end{aligned}$$

Translating the scalar components back to vector notation, where $(a_j c_j) = (\boldsymbol{a} \cdot \boldsymbol{c})$:

$$\boldsymbol{r} = \boldsymbol{b}(\boldsymbol{a} \cdot \boldsymbol{c}) - \boldsymbol{c}(\boldsymbol{a} \cdot \boldsymbol{b})$$

Problem 1.4.1

PROMPT

Given two vectors $\boldsymbol{a}$ and $\boldsymbol{b}$, show that

$$(\boldsymbol{a} \otimes \boldsymbol{a})(\boldsymbol{b} \otimes \boldsymbol{b}) = (\boldsymbol{a} \cdot \boldsymbol{b})\boldsymbol{a} \otimes \boldsymbol{b}$$

SOLUTION

Let the tensor $\boldsymbol{R}$ be the result.

$$\begin{aligned} \boldsymbol{R} = (\boldsymbol{a} \otimes \boldsymbol{a})(\boldsymbol{b} \otimes \boldsymbol{b}) \to R_{ij} &= a_i a_k b_k b_j \\ &= (a_k b_k) a_i b_j \end{aligned}$$

$$R_{ij} = (a_k b_k) a_i b_j \to \boldsymbol{R} = (\boldsymbol{a} \cdot \boldsymbol{b}) \cdot \boldsymbol{a} \otimes \boldsymbol{b}$$

Problem 1.4.2

PROMPT

Find the abstract expression for

$$\frac{\partial}{\partial \boldsymbol{v}}(\boldsymbol{v} \cdot \boldsymbol{v})$$

SOLUTION

$$
\begin{aligned}
\frac{\partial}{\partial \boldsymbol{v}}(\boldsymbol{v} \cdot \boldsymbol{v}) \to \frac{\partial}{\partial \boldsymbol{v}_j}(v_i v_i) &= 2\frac{\partial v_i}{\partial \boldsymbol{v}_j} v_i \\
&= 2\delta_{ij} v_i \\
&= 2v_j
\end{aligned}
$$

Therefore

$$
\frac{\partial}{\partial \boldsymbol{v}}(\boldsymbol{v} \cdot \boldsymbol{v}) = 2\boldsymbol{v}
$$

Problem 1.4.3

PROMPT

Given two second-order tensor $\boldsymbol{A}$ and $\boldsymbol{B}$, show that

$$
\frac{\partial}{\partial \boldsymbol{A}}(\boldsymbol{A} : \boldsymbol{B}) = \boldsymbol{B}
$$

SOLUTION

In index notation:

$$
\frac{\partial}{\partial \boldsymbol{A}}(\boldsymbol{A} : \boldsymbol{B}) \to \frac{\partial}{\partial A_{ij}}(A_{mn} B_{mn})
$$

$$
\begin{aligned}
\frac{\partial}{\partial A_{ij}}(A_{mn} B_{mn}) &= \frac{\partial A_{mn}}{\partial A_{ij}} B_{mn} + \frac{\partial B_{mn}}{\partial A_{ij}} A_{mn} \\
&= \delta_{mi} \delta_{nj} B_{mn} \\
&= B_{ij} \\
&\to \boldsymbol{B}
\end{aligned}
$$

Problem 1.4.4

PROMPT

Find the expression for

$$
\frac{\partial}{\partial \boldsymbol{a}}(\boldsymbol{a} \times \boldsymbol{b})
$$

SOLUTION

In index notation,

$$
\begin{aligned}
\frac{\partial}{\partial \boldsymbol{a}}(\boldsymbol{a} \times \boldsymbol{b}) \to \frac{\partial}{\partial a_l}(\epsilon_{ijk} a_j b_k) &= \epsilon_{ijk} b_k \delta_{jl} \\
&= \epsilon_{ilk} b_k \qquad = -\epsilon_{ikj} b_k
\end{aligned}
$$

Problem 1.4.5

PROMPT

Compute the derivative, with respect to $\boldsymbol{A}$, of

$$
\boldsymbol{A} - \frac{1}{3} \operatorname{trace}(\boldsymbol{A}) \boldsymbol{I}
$$

SOLUTION

$$
\begin{aligned}
\frac{1}{3} \frac{\partial}{\partial A_{mn}}(A_{ij} - A_{kk} \delta_{ij}) &= \delta_{im} \delta_{jn} - \frac{1}{3} \delta_{ij} \delta_{km} \delta_{kn} \\
&= \delta_{im} \delta_{jn} - \frac{1}{3} \delta_{ij} \delta_{mn} \\
&= \mathbb{I}_{ijmn} - \frac{1}{3} \delta_{ij} \delta_{mn} \\
&\to \mathbb{I} - \frac{1}{3} \boldsymbol{I} \otimes \boldsymbol{I}
\end{aligned}
$$

Problem 1.5.1

PROMPT

Using index notation, show that

SOLUTION

a) $\boldsymbol{I} : \boldsymbol{I} = 3$

Since $\boldsymbol{A} : \boldsymbol{B} \to A_{ij} B_{ij}$, then our solution is:

$$
\begin{aligned}
\alpha &= \delta_{ij} \delta_{ij} \\
&= \delta_{ii} \\
&= 3
\end{aligned}
$$

b) $\mathbb{I} : \mathbb{I} = 3\boldsymbol{I}$

By definition, $\mathbb{I}_{ijkl} = \delta_{ik} \delta_{jl}$. Thus

$$
\begin{aligned}
\mathbb{I}_{ijkl}\mathbb{I}_{klmn} &= \big(\delta_{ik}\delta_{jl}\big)(\delta_{km}\delta_{ln}) \\
&= \delta_{im}\delta_{jn} \\
&= \mathbb{I}
\end{aligned}
$$

c) $\mathbb{I}^{\text{sym}} : \mathbb{I}^{\text{sym}} = \mathbb{I}^{\text{sym}}$

$$
\begin{aligned}
\mathbb{I}^{\text{sym}}_{ijmn}\mathbb{I}^{\text{sym}}_{mnkl} &= \frac{1}{2}\big(\delta_{im}\delta_{jn} + \delta_{in}\delta_{jm}\big) \cdot \frac{1}{2}(\delta_{mk}\delta_{nl} + \delta_{ml}\delta_{nk}) \\
&= \frac{1}{4}\big(\delta_{im}\delta_{jn}\delta_{mk}\delta_{nl} + \delta_{im}\delta_{jn}\delta_{ml}\delta_{nk} + \delta_{in}\delta_{jm}\delta_{mk}\delta_{nl} + \delta_{in}\delta_{jm}\delta_{ml}\delta_{nk}\big) \\
&= \frac{1}{4}\big(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} + \delta_{il}\delta_{jk} + \delta_{ik}\delta_{jl}\big) \\
&= \frac{1}{2}\big(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\big) \\
&= \mathbb{I}^{\text{sym}}_{ijkl}
\end{aligned}
$$